

ManyLLMs: Analysis

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1 Overview

This script accompanies the analyses presented in the manuscript. Two versions are provided:

- The `.Rmd` file can be opened and edited in **RStudio**.
- A rendered **PDF** is available for reference, generated by knitting this file.

We provide full R code to ensure the analyses are transparent and reproducible. Where necessary, code is explained or unpacked in-text. This document includes:

1. Visualizations of key variable distributions (used in main text and supplementary materials)
2. Justifications for major analysis decisions

Last updated: 21 May 2025

First, we recoded factor levels and renamed variables in the dataset to improve readability and streamline the subsequent analysis..

1.1 Dataset explanation

Here we summarize the important *Variables* in our dataset.

Our dataset captures each LLM’s response to a moral dilemma in the `opinion` column— a 1–7 *utilitarian score* where higher scores indicate a stronger willingness to endorse the “greater good” at the expense of a moral rule.

The variable `model` identifies the type of *LLMs* which produced this response (e.g., GPT4.1, Llama3, etc) The field `item` (or `example_index`) numbers the *scenario*, and `rep` indicates the *repetition* for each scenario (1, 2, 3..).

The field `Group_Size` codes group size (1 = solo, 2 = pair, 3 = triad).

The variable `type` classifies the moral dilemma (e.g., “Personal”, “Killing–Util”) within the field `dataset` which names the source questionnaire (e.g., `greene`, `oxfor`, `korner_cni`) which are our measures. This is similar to the variable `measurement`.

Finally, the variable **Group** denotes the condition: **Solo** for the *single agents* (baseline of the model) and **Group** for the final group-reflection *consensus*. The variable **step** is the same, but numerical: Solo (**step -1**) vs Group(**step -1**). Note that **Group** indicates both *pairs* and *triads*. **This is our main experimental manipulation.**

We base our analysis on these sets measures to capture **utilitarian boost** in groups of multi-agent LLM settings. Therefore, in our analysis, we always compare **Group** vs. **Solo** condition. You can see a summary of the variables in Table 1.

```
glimpse(combined_dataset)
```

Table 1: Table: Variable Definitions

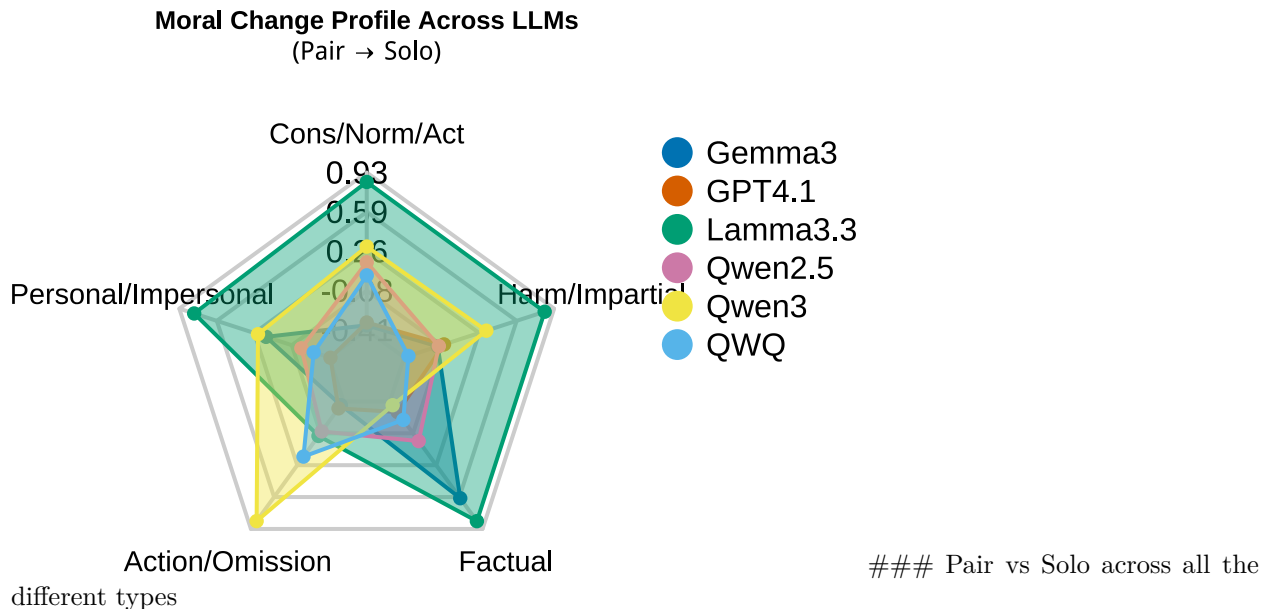
Variable	Description
model	LLM that generated the response (e.g., gpt-4.1, llama3.3)
item	ID of the dilemma scenario - a.k.a expample_index
rep	Repetition index for each scenario (e.g., 1, 2, 3)
group	Solo (baseline) and Group (consensus)
ob/group_size	Group size: n = solo, nn = pair, nnn = triad
type	Moral category of the scenario (e.g., Personal, Impersonal, Killing-Util)
dataset	Measurement (e.g., greene, oxford, korner, cni)
opinion	Utilitarian Score: LLMs' ratings on a 1-7 Likert scale (higher = more utilitarian)

1.2 Moral Chage Across LLMs

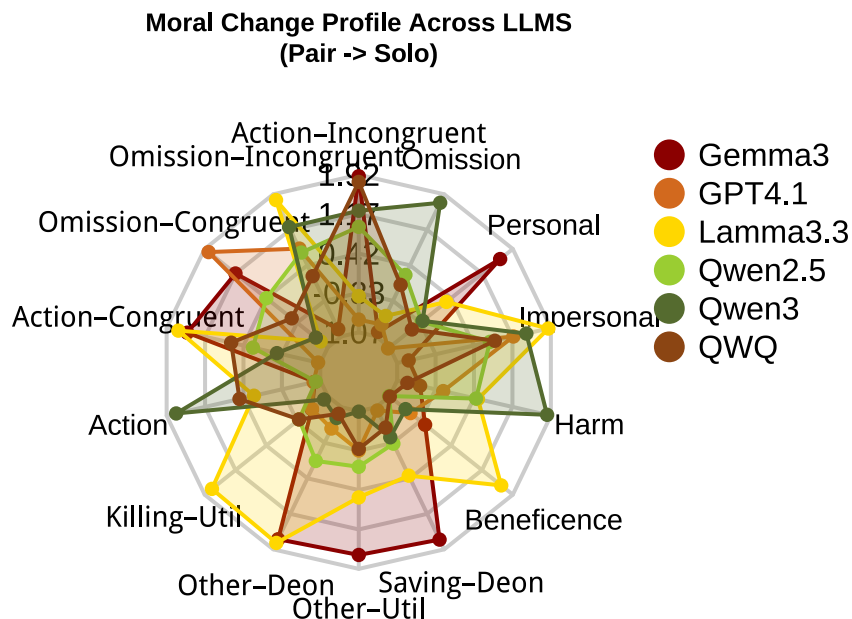
Together our varriables let us track, for each model $\times$ item $\times$ phase $\times$ repetiion, how the LLM's moral judgment shifts from Solo to Group.

1.2.1 Pair vs Solo across all the different measurements

Now We plot Group vs Solo across all the different measurements using an Radar plot.

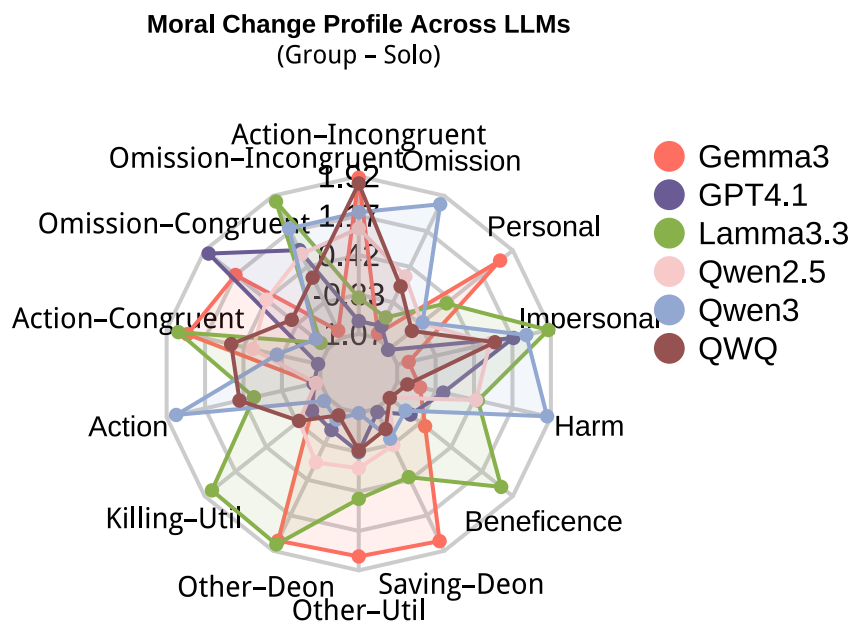


This shows *moral change* profiles across different models for different types. The difference is the Pairs - Solo.



different types

This shows *moral change* profiles across different models. The difference is the Groups (pairs and triads) - Solo.



We see the *profile* of moral change is different across models and types. Next we See each measurement in more depth.

2 Utilitarain Moral Dilemmas

Our **primary reference** is the classic set of *moral utilitarian dilemmas* (‘dataset == “greene”).

```
# Recreate the Greene subset for this document
greene_df <- combined_dataset %>%
  filter(dataset == "greene") %>%
  droplevels()
```

Preparing Data for Ordinal Modeling: We begin by extracting only the Greene subset and converting our response variable to an ordered factor:

```
# Prepare dataset
reflection_moral <- greene_df
reflection_moral$opinion <- as.ordered(reflection_moral$opinion)
```

2.1 Models

The **ordinal** package¹ provides functions for fitting *cumulative link models* (`clm()`) and *cumulative link mixed models* (`clmm()`) to **ordinal** response data. It supports both fixed-effects formulae and random-effects structures, making it ideal for analyzing Likert-type outcomes with clustered or repeated measures.

Next, we fit three candidate cumulative link mixed models to determine the optimal random-effects structure. Each model includes the fixed effect of **Group** (factor type of step) but varies in its random terms:

Note: The variable **rep** denotes **repetition**—each scenario is repeated multiple times to ensure consistency and robustness of the judgments.

```
model1 <- clmm(
  opinion ~ Group + (1 | item),
  data = reflection_moral,
  Hess = TRUE
)
```

1. Model 1: random intercept for each item

This baseline model uses `(1 | item)` to allow every dilemma scenario to have its own starting point on the 1–7 scale. It assumes that the effect of **Group** (single vs. group) is **constant** across all items, but accounts for inherent differences in how “utilitarian” each scenario tends to be.

```
model2 <- clmm(
  opinion ~ Group + (rep | item),
  data = reflection_moral,
  Hess = TRUE
)
```

2. Model 2: random slope of repetitions rep within each item

In addition to item intercepts, this model lets each scenario exhibit its own pattern across repeated presentations. Some items may elicit more consistent ratings across repeats, while others show greater variability, capturing item-specific response stability.

```
model3 <- clmm(
  opinion ~ Group + (1 | item) + (1 | rep),
  data = reflection_moral,
  Hess = TRUE
)
```

¹Christensen, R. H. B. (2019). *ordinal: Regression Models for Ordinal Data*. R package version 2019.12-10. <https://CRAN.R-project.org/package=ordinal>

3. Model 3: crossed random intercepts for `item` and `rep`

This model builds on the previous ones by adding a separate random intercept for each repetitions (`rep`) in addition to the scenario-specific intercepts for each `item`. This accounts for two sources of baseline variation:

- **Item-level:** some dilemmas are judged more utilitarian or deontological on average.
- **Repetition-level:** certain runs (e.g., the first, second, third presentation) may systematically differ.

2.2 Models Comparison and Selection

Model Comparison: with *Likelihood-Ratio* Test: We compare the three fitted `clmm` models using `anova()` to perform a *likelihood-ratio* test. This assesses whether each increase in complexity of the random-effects structure leads to a statistically significant improvement in model fit.

```
anova_res <- anova(model1, model2, model3)
```

Table 2: Table: Likelihood-Ratio Test for Random-Effects Structures

Model	AIC	Chi-square	df	p-value
Model 1	38759.18	NA	NA	NA
Model 3	38761.18	0.00	1	0.9689
Model 2	38652.66	110.52	1	0.0000

2.2.1 Model Selection

Model 2 is the preferred model. The likelihood-ratio test comparing Model 2 to Model 1 yields $p = 0.969$, indicating a significant improvement in fit. Additionally, Model 2 reduces the AIC by 106.53 and the BIC by 91.04 relative to Model 1. These metrics together demonstrate that allowing item-specific repetition effects (Model 2) provides a more parsimonious and better-fitting model.

2.3 Utilitarian Boost

Model 2 was selected as the optimal random-effects structure. Below we summarize its **fixed-effect estimates** and then report **pairwise comparisons** of reflection phases on the probability scale.

We now report the fixed-effect estimates and pairwise comparisons for Model 2 in Table 2.

Table 3: Table: Fixed-Effect Estimates for Model 2

Term	Estimate	Std. Error	z value	p value
1 2	-3.02218	0.39612	-7.62951	0.00000
2 3	-1.04930	0.39530	-2.65446	0.00794
3 4	-0.11148	0.39520	-0.28208	0.77788
4 5	0.45193	0.39525	1.14342	0.25286
5 6	1.48428	0.39546	3.75331	0.00017
6 7	3.68671	0.39665	9.29454	0.00000
GroupSolo	-0.31754	0.04561	-6.96163	0.00000

2.4 Personal vs. Impersonal

Our **primary reference** is the classic set of *moral sacrificial dilemmas* (`dataset == "greene"`), which we analyze across two key dimensions:

- `type == "Personal"` vs. `type == "Impersonal"`

Using our winning random-effects structure, we now test whether the **effect of Group (Group)** differs across **scenario types (type)**. We fit a cumulative link mixed model with an interaction between **Group** and **type**, retaining a random intercept for each dilemma item (**item**).

```
#Fit cumulative link mixed model (CLMM)
model_clmm <- clmm(
  opinion ~ Group * type +
    (1| item),
  data = reflection_moral,
  Hess = TRUE
)

# Summarize the fitted model
summary(model_clmm)

## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: opinion ~ Group * type + (1 | item)
## data:    reflection_moral
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 17049 -19326.86 38673.73 1049(12482) 6.67e-04 3.3e+04
##
## Random effects:
## Groups Name Variance Std.Dev.
## item (Intercept) 11.87 3.446
## Number of groups: item 44
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## GroupSolo 0.10994 0.05946 1.849 0.0645 .
## typePersonal -1.90090 1.05572 -1.801 0.0718 .
## GroupSolo:typePersonal -0.71708 0.07816 -9.175 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
## Estimate Std. Error z value
## 1|2 -3.7103 0.8235 -4.505
## 2|3 -1.7499 0.8232 -2.126
## 3|4 -0.8099 0.8232 -0.984
## 4|5 -0.2456 0.8231 -0.298
## 5|6 0.7877 0.8231 0.957
## 6|7 2.9890 0.8232 3.631
```

We then extract **pairwise estimated marginal means** on the probability scale and present the **contrast** results in a PDF-friendly table.

```
# 4. Obtain pairwise estimated marginal means (on the probability scale)
emms <- emmeans(
  model_clmm,
  pairwise ~ Group * type ,
  type = "response"
)
# 5. View contrast summaries
summary(emms$contrasts)
```

```
## contrast estimate SE df z.ratio p.value
## Group Impersonal - Solo Impersonal -0.110 0.0595 Inf -1.849 0.2505
## Group Impersonal - Group Personal 1.901 1.0600 Inf 1.801 0.2730
## Group Impersonal - Solo Personal 2.508 1.0600 Inf 2.377 0.0816
## Solo Impersonal - Group Personal 2.011 1.0500 Inf 1.907 0.2251
## Solo Impersonal - Solo Personal 2.618 1.0500 Inf 2.484 0.0625
## Group Personal - Solo Personal 0.607 0.0507 Inf 11.969 <.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

P value adjustment: tukey method for comparing a family of 4 estimates

Wihtout rep: (Group-1 Impersonal) - Group7 Impersonal 0.111 0.0594 Inf 1.874 0.2393 (Group-1 Impersonal) - (Group-1 Personal) 2.616 0.9380 Inf 2.789 0.0271 (Group-1 Impersonal) - Group7 Personal 2.021 0.9390 Inf 2.154 0.1364 Group7 Impersonal - (Group-1 Personal) 2.504 0.9390 Inf 2.667 0.0383 Group7 Impersonal - Group7 Personal 1.910 0.9400 Inf 2.033 0.1760 (Group-1 Personal) - Group7 Personal -0.594 0.0507 Inf -11.728 <.0001

Table 4: Contrasts Within Type (Impersonal and Personal)

contrast	estimate	SE	z.ratio	p.value
Group Impersonal - Solo Impersonal	-0.10994	0.05946	-1.84893	0.25046
Group Personal - Solo Personal	0.60714	0.05072	11.96920	0.00000

We see that **Group** opinions are significantly higher than **Solo** opinions across all LLMs for **Personal** dilemmas.

2.5 Pairs and Triads: group size effect.

In our previous models, including a random slope for **repetition** (**rep**) caused convergence failures, so we removed **rep** from the random structure and now focus on how **group size** (**ob**) affects moral judgments.

Because only the **Personal** dilemma type showed a significant interaction with reflection phase, we restrict our analysis to **Personal** scenarios. We then fit a cumulative link mixed model with:

- **Fixed effects:** interaction of **Group** (“Solo” vs. “Group”) and **ob** (n, nn, nnn)
- **Random intercepts:** for each scenario item (**item**)

```
# Fit cumulative link mixed model (CLMM)
model_clmm <- clmm(
  opinion ~ Group * groupsize +
```



```

    (rep| item),
    data = reflection_moral,
    Hess = TRUE
)

# 3. Summarize the fitted model
summary(model_clmm)

```

```

# 4. Obtain pairwise estimated marginal means (on the probability scale)
emms <- emmeans(
  model_clmm,
  pairwise ~ Group * groupsize ,
  type = "response"
)

```

```

# 5. View contrast summaries
summary(emms$contrasts)

```

contrast estimate SE df z.ratio p.value (n Group-1) - nn Group7 -0.712 0.0815 Inf -8.732 <.0001 (n Group-1) - nnn Group7 -0.572 0.0709 Inf -8.061 <.0001 nn Group7 - nnn Group7 0.140 0.0904 Inf 1.546 0.2696

P value adjustment: tukey method for comparing a family of 3 estimates

Table 5: Estimated Marginal Means Contrasts (Tukey-adjusted)

contrast	estimate	SE	df	z.ratio	p.value
Pairs Group - Triads Group	0.096	0.069	Inf	1.394	0.344
Pairs Group - Solo Solo	0.375	0.061	Inf	6.111	0.000
Triads Group - Solo Solo	0.279	0.053	Inf	5.222	0.000

We see that the effect is significant for Solo vs both Group Sizes (pairs and triads). No difference between Pairs and Triads. The plot below show this difference across different items.

2.6 Utilitarian Boost in LLMs

Here we show the effect across each model for the Personal dilemmas. Note that we use `fit_mod <- clmm(opinion ~ Group + (1 | item))` since the `rep/item` does not converge due to negative hessian.

```

# List of models to compare
models_to_compare <- c("Gemma3", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moral %>%
    filter(type == "Personal", model_short == m)
  fit_mod <- clmm(opinion ~ Group + (1 | item), data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ Group, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,

```

```

    Estimate      = estimate,
    `Std. Error`  = SE,
    `z value`     = z.ratio,
    `p value`     = p.value
  )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))

```

Table 6: Table: Group vs. Solo Contrast for Personal Dilemmas Across LLMs

Model	Contrast	Estimate	Std. Error	z value	p value
Gemma3	Group - Solo	1.82342	0.16232	11.23378	0
Lamma3.3	Group - Solo	0.79970	0.15790	5.06449	0
Qwen2.5	Group - Solo	0.67623	0.12362	5.47006	0
Qwen3	Group - Solo	1.14621	0.12970	8.83732	0
QWQ	Group - Solo	0.69064	0.12470	5.53837	0

Since the Hessian for GPT4.1 was not positive definite (i.e. `Error in vcov.clm(...): Hessian is not positive definite`), we instead **pool across both group sizes** and include an interaction term. For GPT4.1

```

reflection_moral_gpt=filter(reflection_moral, model_short=="GPT4.1" & type == "Personal")
fit_mod <- clmm(opinion ~ Group * ob + (1 | item), data = reflection_moral_gpt, Hess = TRUE)
em <- emmeans(fit_mod, pairwise ~ Group * ob, type = "response")

```

Table 7: Estimated Marginal Means for GPT4.1 (Personal) Solo vs Group

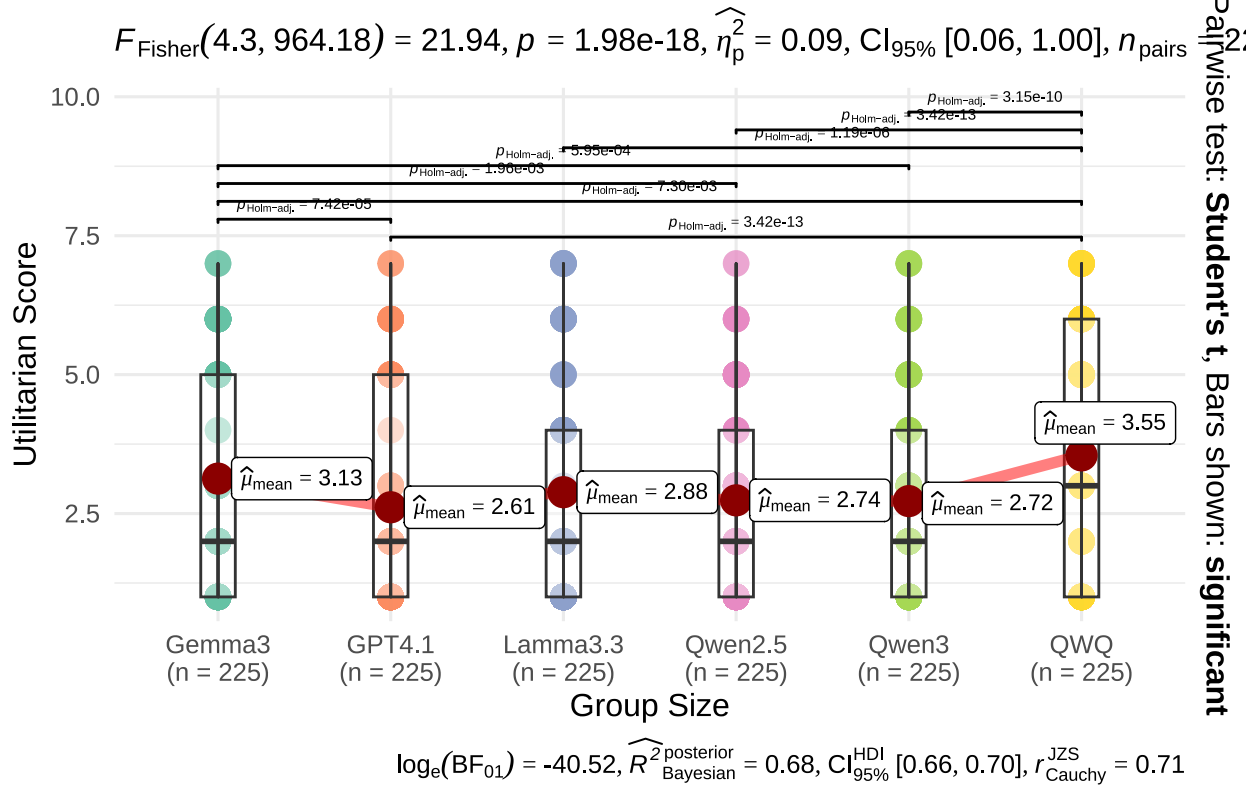
contrast	estimate	SE	df	z.ratio	p.value
nn Group - nnn Group	-0.315	0.240	Inf	-1.313	0.388
nn Group - n Solo	0.250	0.196	Inf	1.273	0.410
nnn Group - n Solo	0.565	0.169	Inf	3.349	0.002

2.7 Baseline and Utilitarian Boost

Here we report some extra analyses on the LLM comparisons, focusing on: **Baseline model of utilitarianism** Examining both the personal dilemmas (`type == "Personal"`) and the overall utilitarian tendency. **Utilitarian shift of the LLMs** - Contrasts between solo and group responses (`Group` effect) across models. **Sanity checks** No difference between agents in `group_size == 2` vs. `group_size == 3`. Sample-balanced version of the analysis using `filter(rep < 15)` to ensure equal representation across group sizes.

2.7.1 Baseline utilitarianism of LLMs

Utilitarian Baseline Across Items (Single-Agent)



Here we fit a cumulative link mixed-effects model (CLMM) to estimate and compare the baseline differences across our “model” factor *different LLMs*, using the Personal condition at the Solo step (`step == -1`). We include a random intercept for each item to account for item-level variability.

```
library(dplyr)
library(ordinal)

# 1) Subset to Personal & Solo (step == -1)
reflection_moral_pb <- reflection_moral %>%
  filter(type == "Personal", step == -1)

# 2) Fit CLMM with random intercepts for item
llm_clmm <- clmm(
  opinion ~ model + (1 | item),
  data = reflection_moral_pb,
  Hess = TRUE
)

summary(llm_clmm)
```

Here we obtain the pairwise estimated marginal means

```
# 4. Obtain pairwise estimated marginal means
emmsmodel <- emmeans(
  llm_clmm,
```

```
pairwise ~ model,
type = "response"
)
```

```
# 5. View contrast summaries
summary(emmsmodel$contrasts)
```

```
## contrast estimate SE df z.ratio p.value
## gemma3:27b - (gpt-4.1) 1.031 0.0872 Inf 11.820 <.0001
## gemma3:27b - llama3.3 0.586 0.0857 Inf 6.836 <.0001
## gemma3:27b - (qwen2.5:32b-instruct) 0.522 0.0845 Inf 6.173 <.0001
## gemma3:27b - qwen3:32b 0.609 0.0853 Inf 7.136 <.0001
## gemma3:27b - qwq -1.103 0.0844 Inf -13.070 <.0001
## (gpt-4.1) - llama3.3 -0.445 0.0863 Inf -5.158 <.0001
## (gpt-4.1) - (qwen2.5:32b-instruct) -0.509 0.0853 Inf -5.969 <.0001
## (gpt-4.1) - qwen3:32b -0.422 0.0858 Inf -4.919 <.0001
## (gpt-4.1) - qwq -2.134 0.0876 Inf -24.359 <.0001
## llama3.3 - (qwen2.5:32b-instruct) -0.064 0.0838 Inf -0.764 0.9735
## llama3.3 - qwen3:32b 0.023 0.0846 Inf 0.272 0.9998
## llama3.3 - qwq -1.688 0.0852 Inf -19.824 <.0001
## (qwen2.5:32b-instruct) - qwen3:32b 0.087 0.0834 Inf 1.043 0.9035
## (qwen2.5:32b-instruct) - qwq -1.624 0.0841 Inf -19.306 <.0001
## qwen3:32b - qwq -1.711 0.0853 Inf -20.071 <.0001
##
## P value adjustment: tukey method for comparing a family of 6 estimates
```

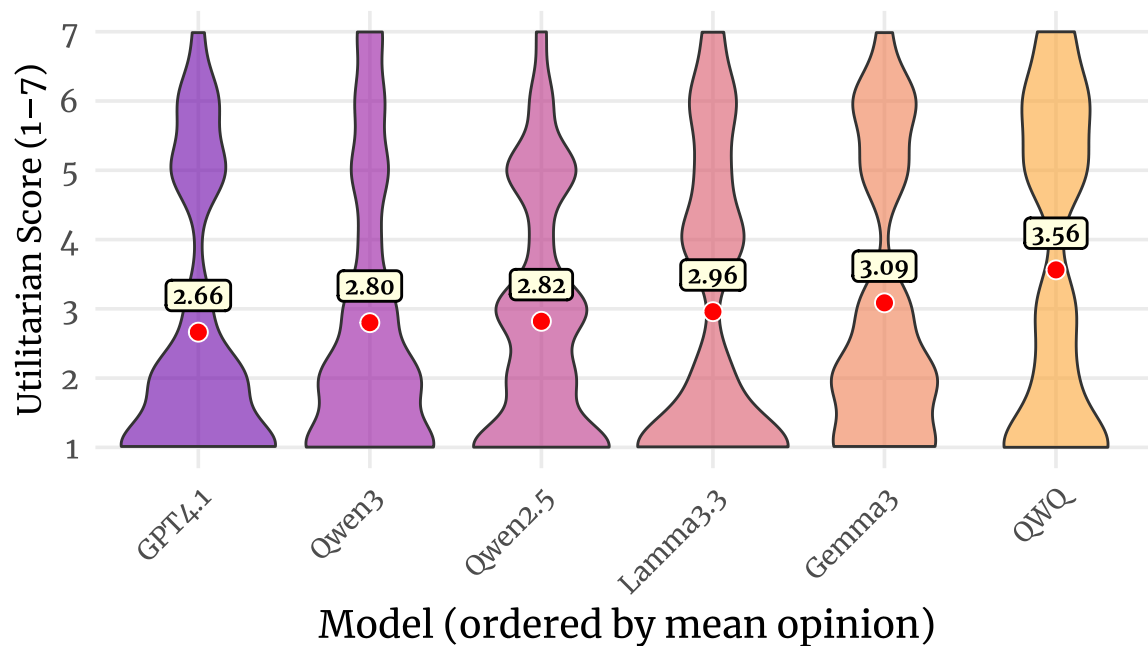
Table 8: Table: Group vs. Solo Contrast for Personal Dilemmas
Across LLMs

contrast	estimate	SE	df	z.ratio	p.value
gemma3:27b - (gpt-4.1)	1.031	0.087	Inf	11.820	0.000
gemma3:27b - llama3.3	0.586	0.086	Inf	6.836	0.000
gemma3:27b - (qwen2.5:32b-instruct)	0.522	0.085	Inf	6.173	0.000
gemma3:27b - qwen3:32b	0.609	0.085	Inf	7.136	0.000
gemma3:27b - qwq	-1.103	0.084	Inf	-13.070	0.000
(gpt-4.1) - llama3.3	-0.445	0.086	Inf	-5.158	0.000
(gpt-4.1) - (qwen2.5:32b-instruct)	-0.509	0.085	Inf	-5.969	0.000
(gpt-4.1) - qwen3:32b	-0.422	0.086	Inf	-4.919	0.000
(gpt-4.1) - qwq	-2.134	0.088	Inf	-24.359	0.000
llama3.3 - (qwen2.5:32b-instruct)	-0.064	0.084	Inf	-0.764	0.973
llama3.3 - qwen3:32b	0.023	0.085	Inf	0.272	1.000
llama3.3 - qwq	-1.688	0.085	Inf	-19.824	0.000
(qwen2.5:32b-instruct) - qwen3:32b	0.087	0.083	Inf	1.043	0.904
(qwen2.5:32b-instruct) - qwq	-1.624	0.084	Inf	-19.306	0.000
qwen3:32b - qwq	-1.711	0.085	Inf	-20.071	0.000

Models are different from each other with respect to baseline. We use `GGStatplot` to show this difference.

Solo Condition: Opinion Distribution by Model

Personal Dilemmas



```
# 0) Font setup
library(showtext)
font_add_google("Merriweather", "merriweather")
showtext_auto()

# 1) Prep data: only Solo (step == -1) & Personal dilemmas, reorder by mean opinion
library(dplyr)
library(forcats)
df_solo <- greene_df %>%
  filter( step == -1) %>%
  mutate(
    model_short = fct_reorder(model_short, opinion, .fun = mean)
  )

# Compute mean per model
means_df <- df_solo %>%
  group_by(model_short) %>%
  summarise(mean_opinion = mean(opinion, na.rm = TRUE), .groups = "drop")

# 2) Plot: violin per model, solo only, with mean labels in a styled box
library(ggplot2)
library(viridis)

ggplot(df_solo, aes(x = model_short, y = opinion, fill = model_short)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_point(
    data = means_df,
    aes(y = mean_opinion),
    color = "white", fill = "red", shape = 21, size = 3
  )
```

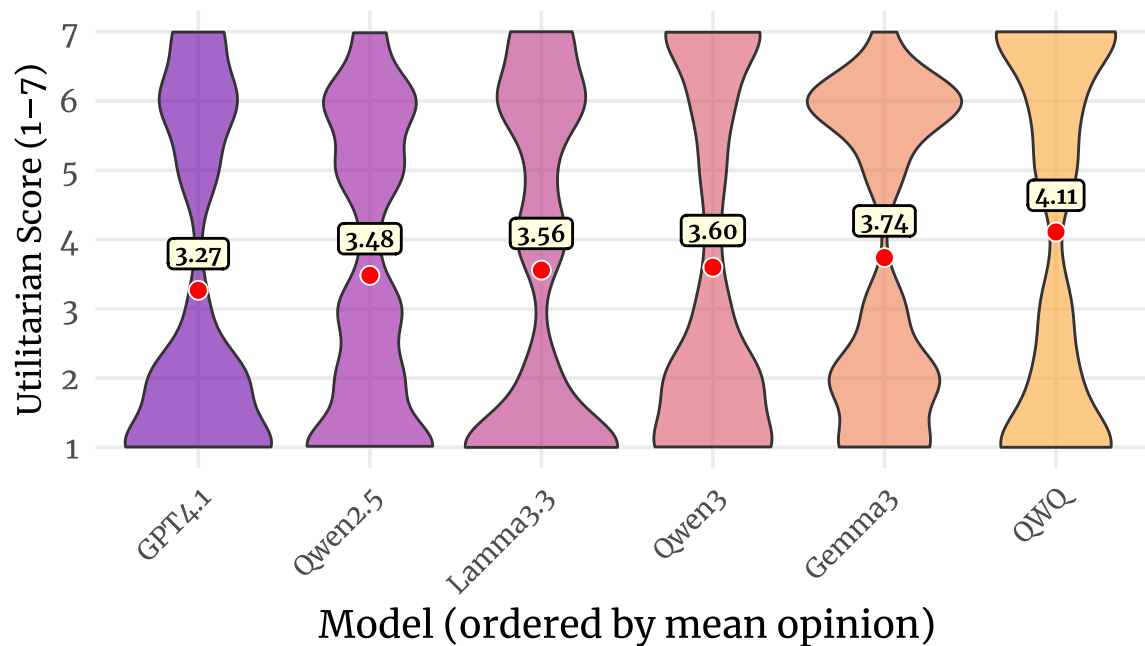
```

) +
geom_label(
  data = means_df,
  aes(y = mean_opinion, label = sprintf("%.2f", mean_opinion)),
  family = "merriweather",
  fontface = "bold",
  size = 3,
  fill = "lightyellow",
  color = "black",
  label.size = 0.5,
  label.r = unit(0.2, "lines"),
  vjust = -0.7
) +
scale_fill_viridis_d(option = "C", begin = 0.2, end = 0.8, guide = "none") +
labs(
  title = "Solo Condition: Opinion Distribution by Model",
  subtitle = "Sacrificial Dilemmas",
  x = "Model (ordered by mean opinion)",
  y = "Utilitarian Score (1-7)"
) +
theme_minimal(base_family = "merriweather", base_size = 14) +
theme(
  plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
  plot.subtitle = element_text(size = 12, hjust = 0.5, margin = margin(b = 10)),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
  axis.title.y = element_text(size = 12),
  panel.grid.minor = element_blank(),
  legend.position = "none",
  plot.margin = margin(20, 20, 20, 20)
) +
scale_y_continuous(limits = c(1, 7), breaks = 1:7)

```

Solo Condition: Opinion Distribution by Model

Sacrificial Dilemmas



Next we show this difference for Impersonal scenarios:

```
# 0) Font setup
font_add_google("Merriweather", "merriweather")
showtext_auto()

# 1) Prep data: only Solo (step == -1) & Personal dilemmas, reorder by mean opinion
library(dplyr)
library(forcats)
df_solo <- greene_df %>%
  filter( type!="Personal", step == -1) %>%
  mutate(
    model_short = fct_reorder(model_short, opinion, .fun = mean)
  )

# Compute mean per model
means_df <- df_solo %>%
  group_by(model_short) %>%
  summarise(mean_opinion = mean(opinion, na.rm = TRUE), .groups = "drop")

# 2) Plot: violin per model, solo only, with mean labels in a styled box
library(ggplot2)
library(viridis)

ggplot(df_solo, aes(x = model_short, y = opinion, fill = model_short)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_point(
    data = means_df,
    aes(y = mean_opinion),
```

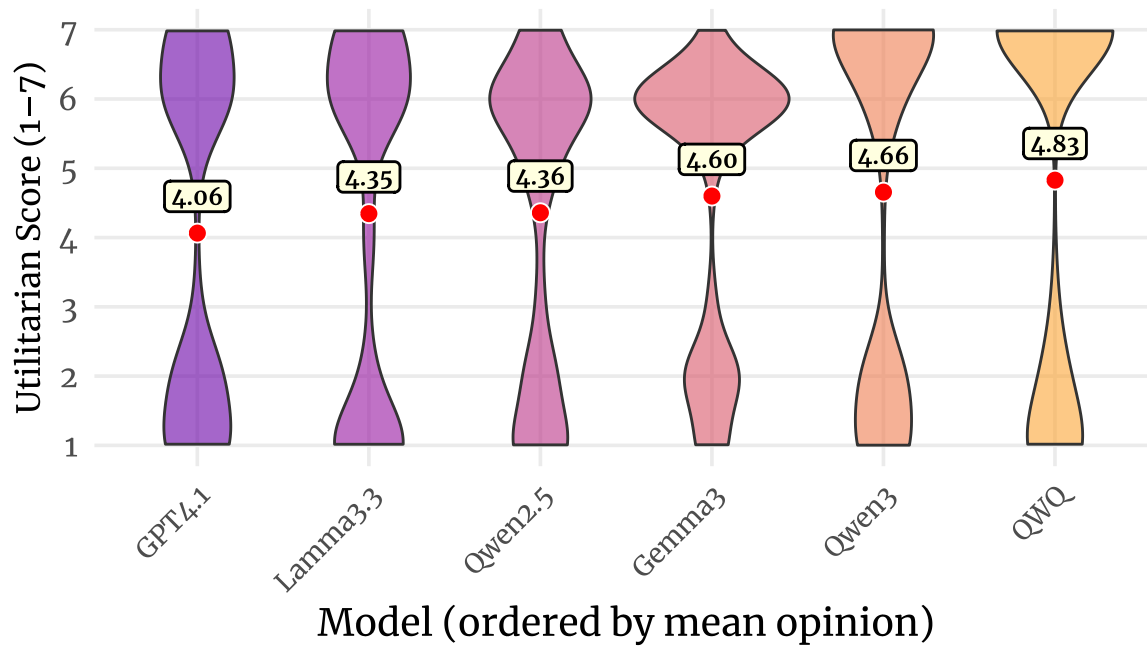
```

    color = "white", fill = "red", shape = 21, size = 3
) +
geom_label(
  data = means_df,
  aes(y = mean_opinion, label = sprintf("%.2f", mean_opinion)),
  family = "merriweather",
  fontface = "bold",
  size = 3,
  fill = "lightyellow",
  color = "black",
  label.size = 0.5,
  label.r = unit(0.2, "lines"),
  vjust = -0.7
) +
scale_fill_viridis_d(option = "C", begin = 0.2, end = 0.8, guide = "none") +
labs(
  title = "Solo Condition: Opinion Distribution by Model",
  subtitle = "Impersonal Dilemmas",
  x = "Model (ordered by mean opinion)",
  y = "Utilitarian Score (1-7)"
) +
theme_minimal(base_family = "merriweather", base_size = 14) +
theme(
  plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
  plot.subtitle = element_text(size = 12, hjust = 0.5, margin = margin(b = 10)),
  axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
  axis.title.y = element_text(size = 12),
  panel.grid.minor = element_blank(),
  legend.position = "none",
  plot.margin = margin(20, 20, 20, 20)
) +
scale_y_continuous(limits = c(1, 7), breaks = 1:7)

```


Solo Condition: Opinion Distribution by Model

Impersonal Dilemmas



2.8 Sanity Checks

2.8.1 Utilitarian score in Personal vs Impersonal:

Personal ratings should be lower than Impersonal scenarios (both LLM and Human prior studies showed that). So we look at the baseline differences for each group. Here we fit a cumulative link mixed-effects model (CLMM) to estimate and compare the differences across our “type” factor *Personal vs Impersonal*, using the both types at the Solo step (`step == -1`).

```
reflection_moral_pb=filter(reflection_moral, step== -1)
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
type_clmm <- clm(
  opinion ~ type,
  data = reflection_moral_pb,
  Hess = TRUE
)

summary(type_clmm)
```

```
## formula: opinion ~ type
## data: reflection_moral_pb
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 13189 -22809.98 45633.97 80(119) 3.62e-04 2.4e+02
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## typePersonal -1.21327 0.03332 -36.41 <2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##      Estimate Std. Error z value
## 1|2 -1.67640    0.03010 -55.701
## 2|3 -0.93622    0.02837 -33.003
## 3|4 -0.63115    0.02793 -22.596
## 4|5 -0.47941    0.02770 -17.306
## 5|6 -0.06001    0.02701  -2.222
## 6|7  1.13985    0.02888  39.473
```

We obtain pairwise estimated marginal means (on the probability scale)

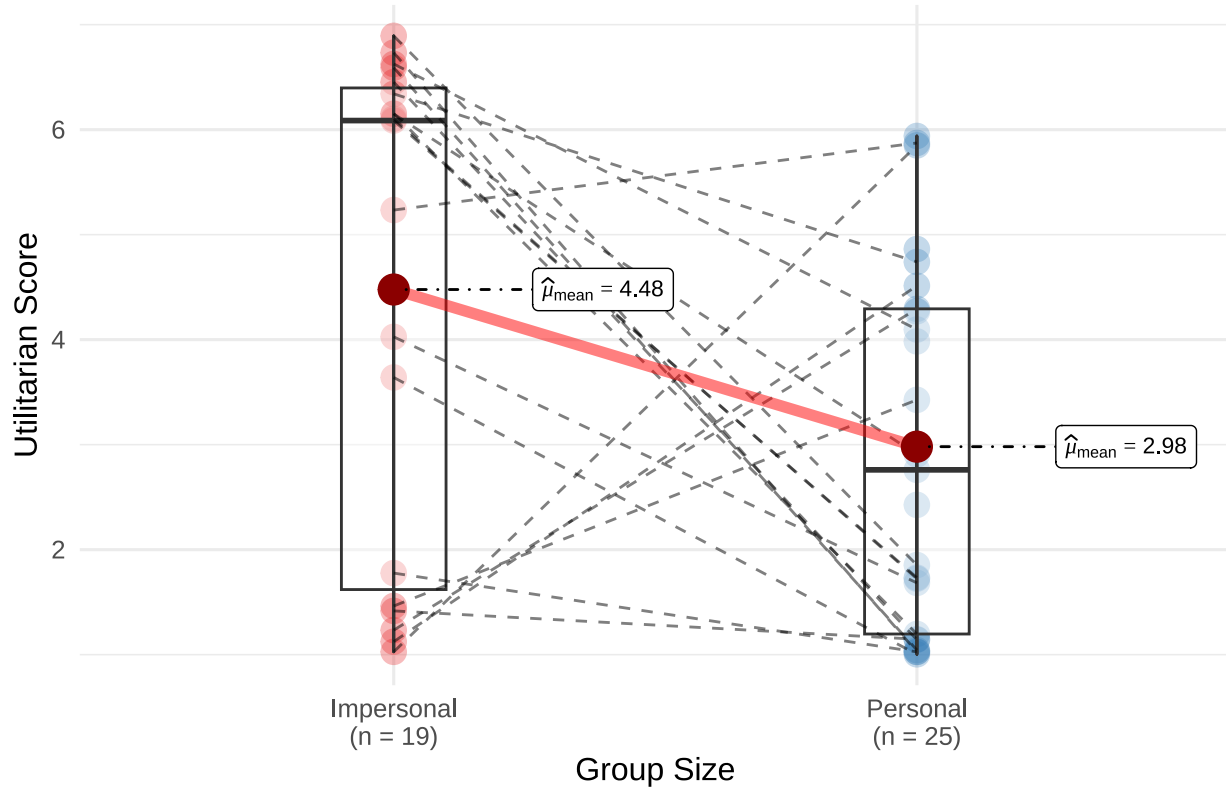
```
# 4. Obtain pairwise estimated marginal means (on the probability scale)
emmstype <- emmeans(
  type_clmm,
  pairwise ~ type,
  type = "response"
)
```

Table 9: Table: Group vs. Solo Contrast for Personal Dilemmas Across LLMs

contrast	estimate	SE	df	z.ratio	p.value
Impersonal - Personal	1.213	0.033	Inf	36.41	0

We see that in general there is a significant difference between the type which can be seen the the plot below:

Utilitarian Baseline Across Types (Single-Agent)



2.8.2 Sanity Check across models

Now we extend this to models. We see the same pattern across all models.

```
reflection_moral_pb=filter(reflection_moral, step==1)
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
# and a random slope of 'step' by example_index:model to mirror (model | example_index)
typellm_clmm <- clmm(
  opinion ~ type * model_short,
  data = reflection_moral_pb,
  Hess = TRUE
)
```

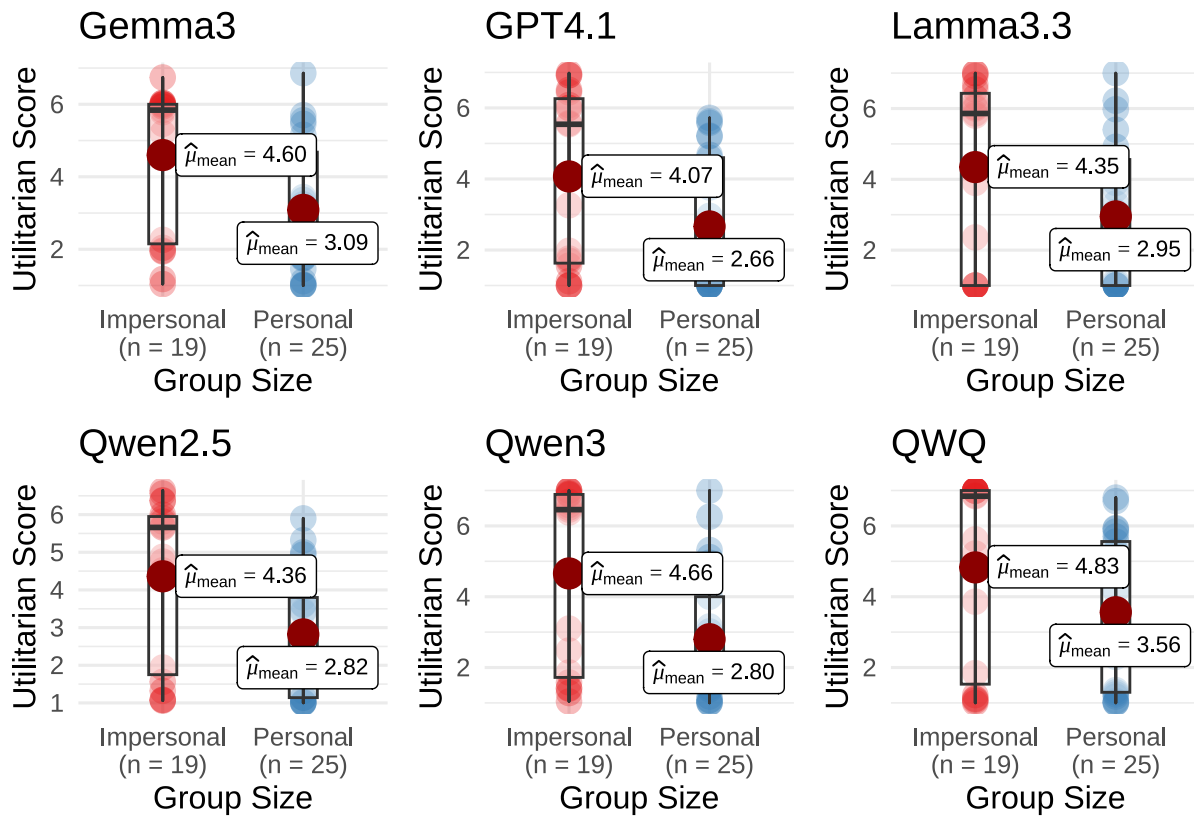
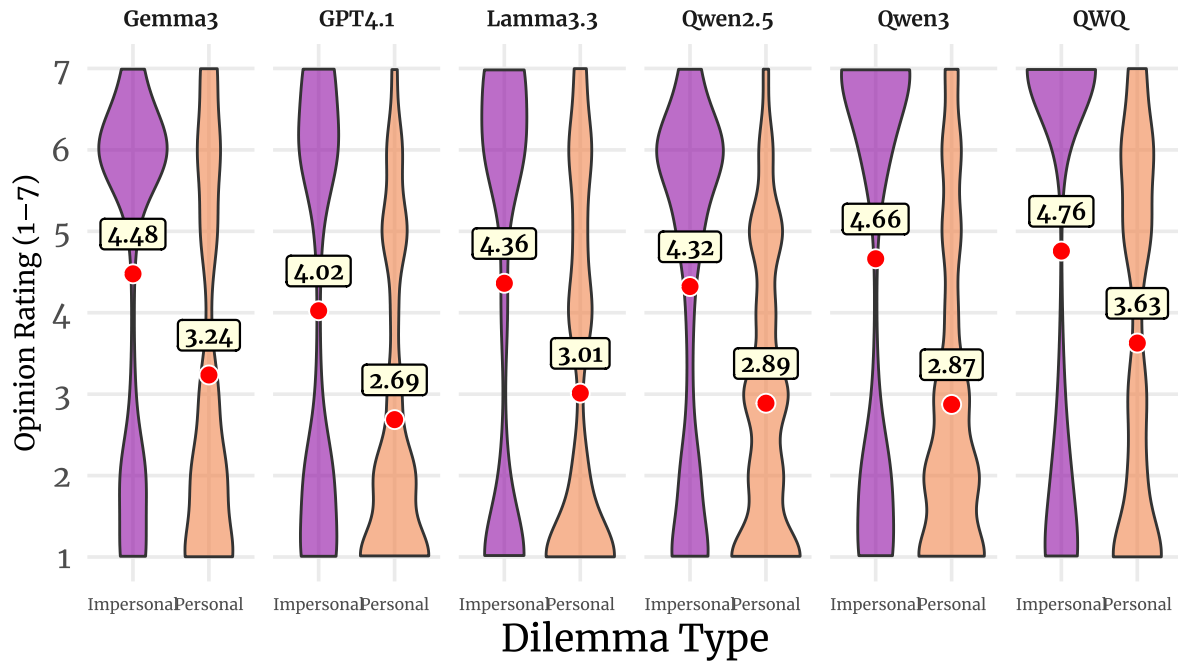
Table 10: Impersonal vs. Personal Contrast (same model)

Contrast	Estimate	Std. Error	DF	z value	p value
Gemma3 Impersonal - Gemma3 Personal	0.9756	0.0718	Inf	13.5822	0
GPT4.1 Impersonal - GPT4.1 Personal	1.1352	0.0792	Inf	14.3347	0
Lamma3.3 Impersonal - Lamma3.3 Personal	1.2078	0.0817	Inf	14.7921	0
Qwen2.5 Impersonal - Qwen2.5 Personal	1.0102	0.0740	Inf	13.6556	0
Qwen3 Impersonal - Qwen3 Personal	1.7214	0.0840	Inf	20.4873	0
QWQ Impersonal - QWQ Personal	1.7369	0.0897	Inf	19.3728	0

The plot shows the Sanity Checks across models.

Opinion Distributions: Impersonal vs. Personal

Personal dilemmas elicit lower utilitarian scores



2.9 Utilitarian Boost Across Models

Plot From Solo to Group: Utilitarian boost across models (we saw that these are all significant)

```
# 0) Font setup
font_add_google("Merriweather", "merriweather")
showtext_auto()

# 1) Prep data
df_violin <- greene_df %>%
  filter(step %in% c(-1, 7), type == "Personal") %>%
  mutate(
    step_label = factor(ifelse(step == -1, "Solo", "Group"),
                        levels = c("Solo", "Group")),
    model_short = factor(model_short)
  )

# 2) Compute means
means_df <- df_violin %>%
  group_by(model_short, step_label) %>%
  summarise(mean_val = mean(opinion, na.rm = TRUE), .groups = "drop")

means_wide <- means_df %>%
  pivot_wider(names_from = step_label, values_from = mean_val)

# 3) Palette with transparency
library(scales)
pal <- c(
  Solo = alpha("#2c7bb6", 0.1),
  Group = alpha("darkgreen", 0.1)
)

# 4) Plot
ggplot(df_violin, aes(step_label, opinion, fill = step_label)) +
  geom_violin(trim = FALSE, alpha = 0.6, color = NA) +
  geom_boxplot(
    width = 0.12, outlier.shape = NA,
    fill = "white", color = "black", alpha = 0.1
  ) +
  geom_jitter(
    aes(color = step_label),
    width = 0.08, size = 1.5,
    alpha = 0.3, show.legend = FALSE
  ) +
  scale_color_manual(values = pal) +
  # connecting lines
  geom_segment(
    data = means_wide, inherit.aes = FALSE,
    aes(x = 1, xend = 2, y = Solo, yend = Group),
    color = "black", size = 0.6
  ) +
  # red point for means
  geom_point(
    data = means_df, inherit.aes = FALSE,
```

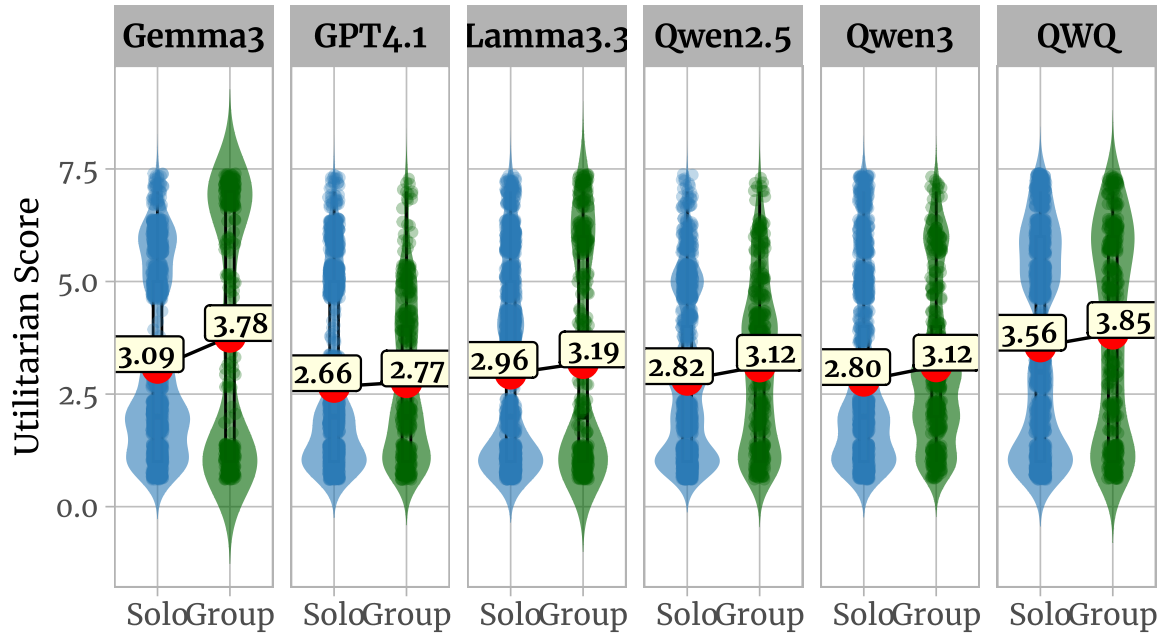
```

    aes(x = step_label, y = mean_val),
    color = "red", size = 5
) +
# callout boxes
geom_label(
  data = means_df, inherit.aes = FALSE,
  aes(
    x = as.numeric(step_label) + ifelse(step_label == "Solo", -0.15, 0.15),
    y = mean_val + 0.3,
    label = sprintf("%.2f", mean_val)
  ),
  family      = "merriweather",
  fontface    = "bold",
  fill        = "lightyellow",
  color       = "black",
  label.size  = 0.4,
  label.r     = unit(0.15, "lines"),
  size        = 3.5
) +
facet_wrap(~ model_short, nrow = 1) +
scale_fill_manual(values = pal) +
labs(
  title      = "Solo vs. Group in Personal Dilemmas",
  subtitle   = "Utilitarian Boost by Model",
  x          = NULL,
  y          = "Utilitarian Score"
) +
theme_light(base_family = "merriweather", base_size = 13) +
theme(
  plot.title      = element_text(size = 18, face = "bold", hjust = 0.5),
  plot.subtitle   = element_text(size = 14, hjust = 0.5, margin = margin(b = 10)),
  strip.text      = element_text(size = 12, face = "bold", color = "black"),
  axis.text.x     = element_text(size = 11),
  axis.title.y    = element_text(size = 12),
  panel.grid.major = element_line(color = "gray"),
  panel.grid.minor = element_blank(),
  legend.position = "none",
  plot.margin     = margin(20, 20, 20, 20)
)

```

Solo vs. Group in Personal Dilemmas

Utilitarian Boost by Model



2.10 Sanity Check: Agent Name

We check whether the agent name would lead to higher/lower Utilitarian Score.

We do this for group size 2 (With two agent)

```
reflection_moral_g=filter(reflection_moral, ob=="nn")
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
# and a random slope of 'step' by example_index:model to mirror (model | example_index)
typellm_clmm <- clmm(
  opinion ~ name + (rep|item) ,
  data = reflection_moral_g,
  Hess = TRUE
)

# 3. Summarize the fitted model
summary(typellm_clmm)
```

We also do this for group size 3 (With three agents).

```
reflection_moral_g=filter(reflection_moral,type=="Personal" & ob=="nnn")
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
# and a random slope of 'step' by example_index:model to mirror (model | example_index)
typellm_clmm <- clmm(
  opinion ~ name + (rep|item) ,
  data = reflection_moral_g,
  Hess = TRUE
)
```

```
# 4. Obtain pairwise estimated marginal means (on the probability scale)
emmstype <- emmeans(
  typellm_clmm,
  pairwise ~ name,
  type = "response"
)
```

Table 11: Estimated Marginal Means Contrasts (Tukey-adjusted)

contrast	estimate	SE	df	z.ratio	p.value
Agent_1 - Agent_2	-0.005	0.13	Inf	-0.037	0.999
Agent_1 - Agent_3	-0.017	0.13	Inf	-0.134	0.990
Agent_2 - Agent_3	-0.013	0.13	Inf	-0.096	0.995

No difference is observed for Agent name (agent 1,2,3). No difference between agents name at stage 7 or Triads.

2.11 LLM Consistency Checks

We want to check the effect of repetition on the the agents responses across all models. We include `rep` to model consistency.

```
reflection_moral_s=filter(reflection_moral,type=="Personal" & ob=="nnn")
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
# and a random slope of 'step' by example_index:model to mirror (1 | example_index)
typellm_clmm <- clmm(
  opinion ~ rep + (1|item) ,
  data = reflection_moral_g,
  Hess = TRUE
)
# 3. Summarize the fitted model
summary(typellm_clmm)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: opinion ~ rep + (1 | item)
## data: reflection_moral_g
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 1328 -1920.78 3857.56 671(4648) 3.38e-04 1.2e+03
##
## Random effects:
## Groups Name Variance Std.Dev.
## item (Intercept) 6.668 2.582
## Number of groups: item 25
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## rep -0.04607 0.06560 -0.702 0.482
##
```



```
## Threshold coefficients:
##      Estimate Std. Error z value
## 1|2  -1.2897    0.5243  -2.460
## 2|3  -0.1529    0.5234  -0.292
## 3|4   0.7295    0.5237   1.393
## 4|5   1.7011    0.5250   3.240
## 5|6   2.2892    0.5262   4.350
## 6|7   3.4733    0.5316   6.533
```

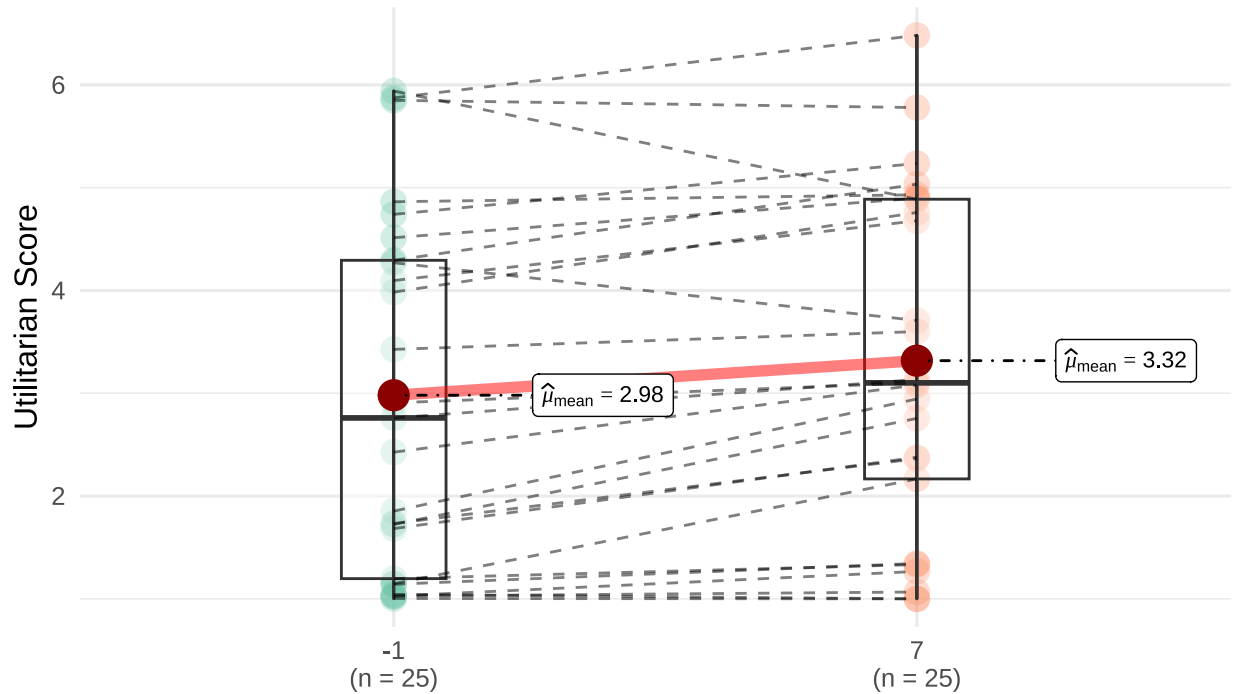
```
reflection_moral_pb=filter(reflection_moral,type=="Personal" & step==7)
# Fit cumulative link mixed model (CLMM) with random intercepts for example_index
# and a random slope of 'step' by example_index:model to mirror (model | example_index)
model_clmm <- clmm(
  opinion ~ rep +
    (1| item),
  data = reflection_moral,
  Hess = TRUE
)
```

No effect of rep is observed.

2.12 Item-based analysis

Here we check how different Items would lead to different utilitarian boost. We plot this change across different types.

Utilitarian Baseline Across Types (Single-Agent)



$\log_e(\text{BF}_{01}) = -2.85$, $\delta_{\text{difference}}^{\text{posterior}} = -0.31$, $\text{CI}_{95\%}^{\text{ETI}} [-0.51, -0.12]$, $r_{\text{Cauchy}}^{\text{JZS}} = 0.71$

First we create a lookup table of the scenarios

```
dilemma_lookup <- data.frame(
  example_index = c(21:39, 40:64),
  label = c(
    paste0(1:19, c(
      "Standard Trolley", "Standard Fumes", "Donation", "Transplant (Impersonal)", "Hospital Swap",
      "Minefield", "Footbridge Switch", "Bomb Shelter", "Back Alley", "Dog Dilemma",
      "Accidental Bump", "Oven", "Journalist", "Crash", "Sinkhole",
      "Missing Bridge", "Driver Switch", "Emergency Plan", "Eyes"
    )),
    paste0(1:24, c(
      "Transplant", "Footbridge", "Country Road", "Boss", "Lifeboat",
      "Hard Times", "Smother for Dollars", "Safari", "Crying Baby", "Plane Crash",
      "Hired Rapist", "Grandson", "Infanticide", "Preventing the Spread", "Modified Lifeboat",
      "Modified Preventing the Spread", "Modified Safari", "Modified Bomb", "Submarine",
      "Lawrence of Arabia", "Sophie's Choice", "Sacrifice", "Vitamins", "Vaccine Test", "Euthanasia"
    ))
  )
)
```

```
dilemma_lookup$example_index=as.factor(dilemma_lookup$example_index)
```

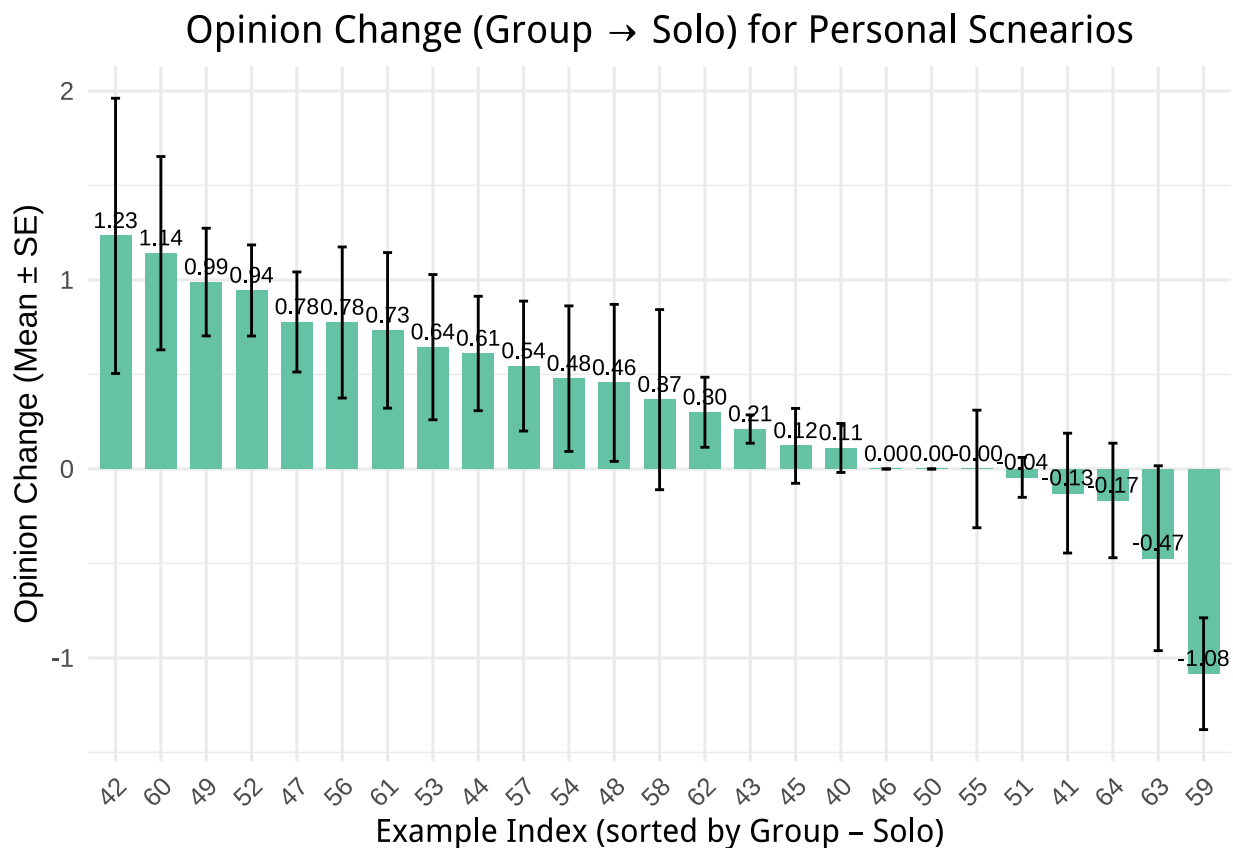
```
combined_dataset$example_index=as.factor(combined_dataset$example_index)
# Step 1: Compute difference scores and SD
df_diff_all <- combined_dataset %>%
  filter(dataset=="green") %>%
  group_by(example_index, rep, model, step, type) %>%
  summarise(mean_opinion = mean(opinion, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = step,
    values_from = mean_opinion,
    names_prefix = "step_"
  ) %>%
  filter(!is.na(`step_1`) & !is.na(`step_7`)) %>%
  mutate(
    diff = `step_7` - `step_1`,
    abs_diff = abs(diff)
  ) %>%
  group_by(example_index, type) %>%
  summarise(
    mean_diff = mean(diff, na.rm = TRUE),
    sd_diff = sd(diff, na.rm = TRUE),
    n = n(),
    se_diff = sd_diff / sqrt(n),
    .groups = "drop"
  ) %>%
  arrange(desc(mean_diff)) %>%
  mutate(example_index = factor(example_index, levels = example_index))

df_diff_allp=filter(df_diff_all,type=="Personal")
# Step 2: Plot with error bars
ggplot(df_diff_allp, aes(x = example_index, y = mean_diff)) +
  geom_col(fill = "#66c2a5", width = 0.7) +
  geom_errorbar(aes(ymin = mean_diff - se_diff, ymax = mean_diff + se_diff),
```

```

width = 0.2, color = "black") +
geom_text(aes(label = sprintf("%.2f", mean_diff)),
          vjust = -0.5, size = 3) +
labs(
  x = "Example Index (sorted by Group - Solo)",
  y = "Opinion Change (Mean ± SE)",
  title = "Opinion Change (Group → Solo) for Personal Scenarios"
) +
theme_minimal(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),
  plot.title = element_text(hjust = 0.5, face = "bold")
)

```



```

dilemma_lookup$example_index=as.factor(dilemma_lookup$example_index)

df_diff_all$example_index=as.factor(df_diff_all$example_index)
df_plot <- df_diff_all %>%
  left_join(dilemma_lookup, by = "example_index") %>%
  arrange(desc(mean_diff)) %>%
  mutate(label = factor(example_index, levels = example_index))

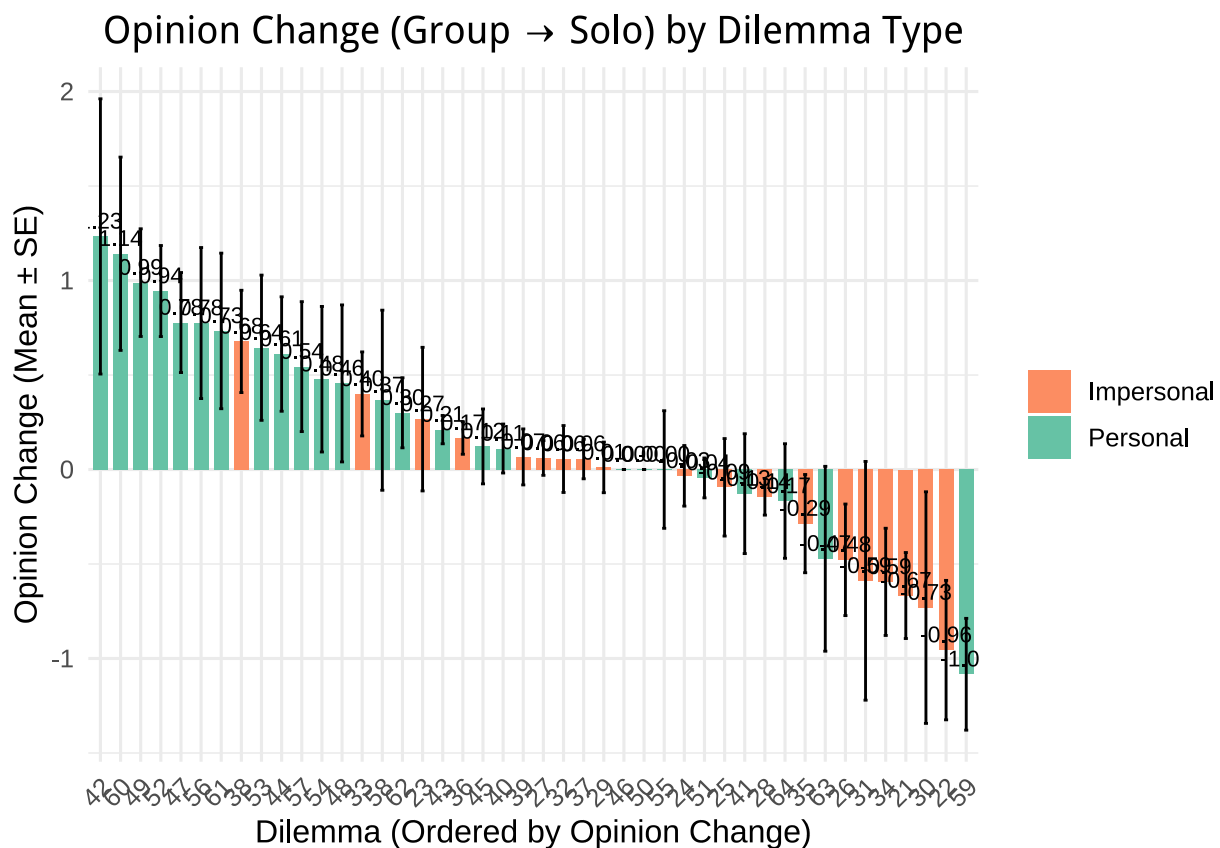
# Plot
ggplot(df_plot, aes(x = label, y = mean_diff, fill = type)) +
  geom_col(width = 0.7) +
  geom_errorbar(aes(ymin = mean_diff - se_diff, ymax = mean_diff + se_diff),

```

```

width = 0.2, color = "black") +
geom_text(aes(label = sprintf("%.2f", mean_diff)), vjust = -0.5, size = 3) +
labs(
  x = "Dilemma (Ordered by Opinion Change)",
  y = "Opinion Change (Mean ± SE)",
  title = "Opinion Change (Group → Solo) by Dilemma Type"
) +
scale_fill_manual(values = c("Personal" = "#66c2a5", "Impersonal" = "#fc8d62")) +
theme_minimal(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),
  plot.title = element_text(hjust = 0.5, face = "bold"),
  legend.title = element_blank()
)

```



2.13 Balanced Sample checks

Here we include the `rep<16` so we have a balanced sample and check the robustness of the result

```

reflection_moralrep=filter(reflection_moral, rep<16)
model2rep <- clmm(
  opinion ~ Group *type + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

```

Similar to the complete sample, we see the effect of Utilitarian boost in personal moral scenarios.

4. Obtain pairwise estimated marginal means (on the probability scale)

```
emms <- emmeans(
  model2rep,
  pairwise ~ Group * type ,
  type = "response"
)
emms
```

```
## $emmeans
##   Group type      emmean    SE  df asymp.LCL asymp.UCL
##   Group Impersonal  0.418 0.632 Inf   -0.821    1.6572
##   Solo  Impersonal  0.476 0.655 Inf   -0.809    1.7603
##   Group Personal   -1.169 0.556 Inf   -2.258   -0.0795
##   Solo  Personal   -1.762 0.574 Inf   -2.887   -0.6368
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast              estimate    SE  df z.ratio p.value
##   Group Impersonal - Solo Impersonal -0.0578 0.0840 Inf   -0.687  0.9020
##   Group Impersonal - Group Personal  1.5866 0.8410 Inf    1.887  0.2338
##   Group Impersonal - Solo Personal  2.1799 0.8530 Inf    2.554  0.0520
##   Solo Impersonal - Group Personal  1.6443 0.8590 Inf    1.915  0.2217
##   Solo Impersonal - Solo Personal  2.2376 0.8710 Inf    2.569  0.0500
##   Group Personal - Solo Personal  0.5933 0.0726 Inf    8.174 <.0001
##
## P value adjustment: tukey method for comparing a family of 4 estimates
```

2.14 Action vs. Omission

2.14.1 Action

We use the same model as before, but here we focus on the `type == "Action"` vs. `type == "Omission"`

```
reflection_moralrep=filter(combined_dataset, dataset=="keshmirian" & type=="Action")
reflection_moralrep$opinion=as.factor(reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)
```

We do not see any effect of Action. We investigate this effect for each models.

```
emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms
```

```
## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1      0.801 0.579 Inf   -0.333     1.94
##    7      0.972 0.576 Inf   -0.156     2.10
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7  -0.171 0.0958 Inf  -1.783  0.0746

# List of models to compare
models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts
```

```
##      Model      Contrast Estimate Std. Error z value p value
## 1 Qwen2.5 (stepf-1) - stepf7  0.7300191  0.1842676  3.961733 7.440769e-05
## 2   Qwen3 (stepf-1) - stepf7 -0.9473348  0.1917448 -4.940601 7.788199e-07
## 3    QWQ (stepf-1) - stepf7 -0.6542610  0.2000762 -3.270059 1.075251e-03
```

2.14.2 Omission

We use the same model as before, but here we focus type == "Omission"

```
reflection_moralrep=filter(combined_dataset, dataset=="keshmirian" & type=="Omission")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
```

```

data = reflection_moralrep,
Hess = TRUE
)

```

We do not see any effect of Action. We investigate this effect for each models.

```

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

```

```

## $emmeans
##      stepf emmean      SE df asymp.LCL asymp.UCL
##    -1      -1.07 0.998 Inf      -3.03      0.888
##     7      -1.15 1.010 Inf      -3.13      0.821
##
## Confidence level used: 0.95
##
## $contrasts
##      contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7      0.086 0.102 Inf   0.845  0.3984

```

```

# List of models to compare
models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

```

```

##      Model      Contrast      Estimate Std. Error      z value      p value

```

```
## 1 Gemma3 (stepf-1) - stepf7  0.8680338  0.2507668  3.461518 0.0005371378
## 2 Qwen3 (stepf-1) - stepf7 -0.4533551  0.1828191 -2.479801 0.0131455672
```

3 Factual Utilitarian Dilemma (dataset: Korner)

3.1 Killing - Utilitarian

We use the same model as before, but here we focus `type == "Killing-Util"`

```
reflection_moralrep=filter(combined_dataset, dataset=="korner" & type=="Killing-Util")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms
```

```
## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1      1.47 0.465 Inf      0.558      2.38
##    7      1.60 0.466 Inf      0.687      2.52
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.133 0.0702 Inf  -1.889  0.0588
```

```
models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast   = contrast,
      Estimate   = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}
```



```

    )
  }
  contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
  # Filter to only the contrasts with p value below 0.05
  significant_contrasts <- contrast_tbl %>%
    filter(`p value` < 0.05)

  # Display them
  significant_contrasts

```

```

##      Model      Contrast      Estimate Std. Error    z value      p value
## 1  Gemma3 (stepf-1) - stepf7 -1.9460706  0.2052745  -9.480334 2.534716e-21
## 2   GPT4.1 (stepf-1) - stepf7  0.2870755  0.1273634   2.253988 2.419691e-02
## 3 Lamma3.3 (stepf-1) - stepf7 -1.0986425  0.1501942  -7.314813 2.577407e-13
## 4   Qwen2.5 (stepf-1) - stepf7  0.5013118  0.1306587   3.836802 1.246467e-04

```

3.2 Killing - Utilitarian

We use the same model as before, but here we focus type == "Killing-Util"

```

reflection_moralrep=filter(combined_dataset, dataset=="korner" & type=="Other-Util")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

```

```

## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1    -0.784 0.654 Inf     -2.07     0.497
##    7    -0.501 0.660 Inf     -1.80     0.792
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.283 0.0839 Inf  -3.369  0.0008

```

```

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%

```

```

filter(model_short == m)

fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
as.data.frame(summary(em$contrasts)) %>%
  transmute(
    Model      = m,
    Contrast    = contrast,
    Estimate    = estimate,
    `Std. Error` = SE,
    `z value`   = z.ratio,
    `p value`   = p.value
  )
}
contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

```

```

##      Model      Contrast  Estimate Std. Error  z value    p value
## 1  GPT4.1 (stepf-1) - stepf7 -0.4376896  0.1637419 -2.673046 0.0075165984
## 2  Lamma3.3 (stepf-1) - stepf7 -0.7210882  0.1859937 -3.876949 0.0001057746

```

We use the same model as before, but here we focus `type == "Other-Deon"`

3.3 Other-Deontology

```

reflection_moralrep=filter(combined_dataset, dataset=="korner" & type=="Other-Deon")
reflection_moralrep$opinion=as.factor(reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

```

```

## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1      1.48 0.539 Inf    0.4271     2.54
##    7      1.18 0.578 Inf    0.0459     2.31
##

```

```
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df z.ratio p.value
## (stepf-1) - stepf7 0.304 0.104 Inf 2.918 0.0035

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model = m,
      Contrast = contrast,
      Estimate = estimate,
      `Std. Error` = SE,
      `z value` = z.ratio,
      `p value` = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts
```

```
##      Model      Contrast Estimate Std. Error z value p value
## 1  Gemma3 (stepf-1) - stepf7 -1.5651464 0.3371015 -4.642953 3.434638e-06
## 2  GPT4.1 (stepf-1) - stepf7 0.9322985 0.2134915 4.366911 1.260158e-05
## 3 Lamma3.3 (stepf-1) - stepf7 -2.7710006 0.3061397 -9.051425 1.411147e-19
## 4  Qwen2.5 (stepf-1) - stepf7 0.4529952 0.2107097 2.149854 3.156674e-02
## 5   Qwen3 (stepf-1) - stepf7 1.2118074 0.2184515 5.547262 2.901785e-08
## 6    QWQ (stepf-1) - stepf7 0.7999014 0.2397394 3.336546 8.482642e-04
```

We use the same model as before, but here we focus `type == "Saving-Deon"`

3.4 Saving-Deontology

```
reflection_moralrep=filter(combined_dataset, dataset=="korner" & type=="Saving-Deon")
reflection_moralrep$opinion=as.factor(reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
```

```

)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1      0.265 0.327 Inf   -0.376    0.905
##    7      0.280 0.353 Inf   -0.411    0.972
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast          estimate    SE df z.ratio p.value
## (stepf-1) - stepf7 -0.0157 0.108 Inf  -0.146  0.8841

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

##      Model      Contrast  Estimate Std. Error  z value    p value
## 1  Gemma3 (stepf-1) - stepf7 -1.2437735  0.3175172 -3.917185 8.958903e-05
## 2  GPT4.1 (stepf-1) - stepf7  0.7476768  0.2097010  3.565443 3.632426e-04
## 3 Lamma3.3 (stepf-1) - stepf7 -1.0107099  0.2480173 -4.075159 4.598284e-05

```

4 Oxford Utilitarian Scale (dataset: oxford)

4.1 Instrumental Harm

```
reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Harm")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##      stepf emmean      SE df asymp.LCL asymp.UCL
##    -1      -2.23 0.725 Inf      -3.65      -0.812
##     7      -3.32 0.771 Inf      -4.84      -1.813
##
## Confidence level used: 0.95
##
## $contrasts
##      contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7      1.09 0.139 Inf   7.849  <.0001

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
```

```

filter(`p value` < 0.05)

# Display them
significant_contrasts

##      Model          Contrast Estimate Std. Error z value      p value
## 1  Gemma3 (stepf-1) - stepf7 3.2100217  0.4158041  7.720034 1.162989e-14
## 2  GPT4.1 (stepf-1) - stepf7 0.6233662  0.2881983  2.162977 3.054299e-02
## 3 Lamma3.3 (stepf-1) - stepf7 1.4613199  0.3281874  4.452699 8.479747e-06
## 4  Qwen2.5 (stepf-1) - stepf7 0.7205419  0.3009903  2.393904 1.667010e-02
## 5      QWQ (stepf-1) - stepf7 1.2087977  0.2850573  4.240543 2.229795e-05

```

4.2 Impartial Beneficence

```

reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Beneficence")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

```

```

## $emmeans
##   stepf emmean    SE df asymp.LCL asymp.UCL
##   -1      0.413 0.556 Inf   -0.6770     1.50
##    7      1.131 0.557 Inf    0.0399     2.22
##
## Confidence level used: 0.95
##
## $contrasts
##   contrast          estimate    SE df z.ratio p.value
## (stepf-1) - stepf7   -0.718 0.117 Inf  -6.123  <.0001

```

```

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%

```

```

    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
  }
  contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
  # Filter to only the contrasts with p value below 0.05
  significant_contrasts <- contrast_tbl %>%
    filter(`p value` < 0.05)

  # Display them
  significant_contrasts

##      Model      Contrast      Estimate Std. Error  z value      p value
## 1  GPT4.1 (stepf-1) - stepf7 -0.5716165  0.2327300 -2.456136 1.404400e-02
## 2 Lamma3.3 (stepf-1) - stepf7 -3.1830955  0.3366475 -9.455278 3.221661e-21

```

5 CNI Utilitarian Dilemmas (dataset: CNI)

5.1 Action-Incongruent

```

reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Beneficence")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##      stepf emmean      SE df asymp.LCL asymp.UCL
## -1      0.413 0.556 Inf   -0.6770      1.50
## 7      1.131 0.557 Inf    0.0399      2.22
##
## Confidence level used: 0.95
##
## $contrasts
##      contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.718 0.117 Inf  -6.123  <.0001

```

```

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast   = contrast,
      Estimate   = estimate,
      `Std. Error` = SE,
      `z value`  = z.ratio,
      `p value`  = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

```

```

##      Model      Contrast      Estimate Std. Error  z value      p value
## 1  GPT4.1 (stepf-1) - stepf7 -0.5716165  0.2327300 -2.456136 1.404400e-02
## 2 Lamma3.3 (stepf-1) - stepf7 -3.1830955  0.3366475 -9.455278 3.221661e-21

```

5.2 Omission–Incongruent

```

reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Beneficence")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##      stepf emmean    SE df asymp.LCL asymp.UCL
##      -1      0.413 0.556 Inf   -0.6770      1.50

```



```
## 7      1.131 0.557 Inf      0.0399      2.22
##
## Confidence level used: 0.95
##
## $contrasts
## contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.718 0.117 Inf   -6.123  <.0001
```

```
models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts
```

```
##      Model      Contrast      Estimate Std. Error  z value      p value
## 1   GPT4.1 (stepf-1) - stepf7 -0.5716165  0.2327300 -2.456136 1.404400e-02
## 2 Lamma3.3 (stepf-1) - stepf7 -3.1830955  0.3366475 -9.455278 3.221661e-21
```

5.3 Action–Congruent”

```
reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Beneficence")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)
model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
```

```

pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##      stepf emmean      SE df asymp.LCL asymp.UCL
## -1      0.413 0.556 Inf   -0.6770      1.50
##  7      1.131 0.557 Inf    0.0399      2.22
##
## Confidence level used: 0.95
##
## $contrasts
##      contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.718 0.117 Inf  -6.123  <.0001

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

```

```

##      Model      Contrast      Estimate Std. Error  z value      p value
## 1  GPT4.1 (stepf-1) - stepf7 -0.5716165  0.2327300 -2.456136 1.404400e-02
## 2 Lamma3.3 (stepf-1) - stepf7 -3.1830955  0.3366475 -9.455278 3.221661e-21

```

5.4 Omission–Congruent”

```

reflection_moralrep=filter(combined_dataset, dataset=="oxford" & type=="Beneficence")
reflection_moralrep$opinion=as.factor (reflection_moralrep$opinion)

```

```

model2rep <- clmm(
  opinion ~ stepf + (rep | item),
  data = reflection_moralrep,
  Hess = TRUE
)

emms <- emmeans(
  model2rep,
  pairwise ~ stepf ,
  type = "response"
)
emms

## $emmeans
##      stepf emmean      SE df asymp.LCL asymp.UCL
##    -1      0.413 0.556 Inf   -0.6770      1.50
##     7      1.131 0.557 Inf    0.0399      2.22
##
## Confidence level used: 0.95
##
## $contrasts
##      contrast      estimate      SE df z.ratio p.value
## (stepf-1) - stepf7   -0.718 0.117 Inf  -6.123 <.0001

models_to_compare <- c("Gemma3", "GPT4.1", "Lamma3.3", "Qwen2.5", "Qwen3", "QWQ")

# Function to fit the model and extract contrasts
get_contrast <- function(m) {
  df_mod <- reflection_moralrep %>%
    filter(model_short == m)

  fit_mod <- clm(opinion ~ stepf, data = df_mod, Hess = TRUE)
  em <- emmeans(fit_mod, pairwise ~ stepf, type = "response")
  as.data.frame(summary(em$contrasts)) %>%
    transmute(
      Model      = m,
      Contrast    = contrast,
      Estimate    = estimate,
      `Std. Error` = SE,
      `z value`   = z.ratio,
      `p value`   = p.value
    )
}

contrast_tbl <- bind_rows(lapply(models_to_compare, get_contrast))
# Filter to only the contrasts with p value below 0.05
significant_contrasts <- contrast_tbl %>%
  filter(`p value` < 0.05)

# Display them
significant_contrasts

##      Model      Contrast      Estimate Std. Error  z value      p value
## 1  GPT4.1 (stepf-1) - stepf7 -0.5716165  0.2327300 -2.456136 1.404400e-02

```

```
## 2 Lamma3.3 (stepf-1) - stepf7 -3.1830955 0.3366475 -9.455278 3.221661e-21
```